

EBN111 SEMESTER TEST / TOETS (MAR 2017)**ANSWER SHEET**

Surname: Van:		Student number: Studente nommer:							
Initials: Voorletters:		English (Eng 3-1)	English (Eng 3-6)	Afrikaans					

SECTION / AFDELING 1 (18)

01	$R = 5.80$	02	$i = 25 \text{ A}$
03	$P_{(ind \text{ } V\text{-source})} = 12 \text{ W}$	04	$P_{(dep \text{ } I\text{-source})} = 10 \text{ W}$
05	$R = 90 \Omega$		
06	$R_{ab} = 40 \Omega$		
07	$R_1 = 25 \text{ k}\Omega$	08	$R_2 = 6.25 \text{ k}\Omega$
09	$I_S = -8 \text{ A}$	10	$V_1 = 15.625 \text{ V}$

SECTION / AFDELING 2 (20)

11	$\theta = -4$	12	$\sigma = -2$
13	$a = -350$	14	$b = 150$
15	$c = 1$	16	$d = 1$
17	$e = 10$	18	$f = -2$
19	$g = -1$	20	$h = 4$
21	$k = 0.12$	22	$l = -0.1$
23	$p = 0.125$	24	$q = 10 \text{ mA}$
25	$V_0 = -60 \text{ V}$		
26	$V_{CE} = 5.1 \text{ V}$	27	$R_B = 50 \text{ k}\Omega$

QUESTION 3 (8)

28	$I_{Y(4A)} = 800 \text{ mA}$	29	$I_{Y(10V)} = 400 \text{ mA}$
30	$V_{TH} = 10 \text{ V}$	31	$R_{TH} = 4 \text{ k}\Omega$